NEUROFLIGHT: EMG-CONTROLLED DRONE NAVIGATION USING NPG LITE BIOSIGNAL ACQUISITION AND LITEWING ESP32-S3

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Abstract

Conventional drones are flown with joysticks, buttons or touchscreen applications, which can be inconvenient for some users. This paper presents NeuroFlight, a low-cost prototype that uses surface electromyography (EMG) from the left and right forearms as a hands-free control input for a drone. An NPG Lite biosignal device streams two EMG channels at 500 Hz over Bluetooth Low Energy (BLE) to a Python application on a laptop. Each channel is passed through a causal 20–100 Hz Butterworth band-pass filter and a 50 Hz notch filter, and a 50 ms sliding-window RMS envelope is compared with a threshold learned during a three-second relaxed-muscle calibration. Activity is classified as left, right, both (treated as STOP) or none and shown on a live Tkinter dashboard. Commands are sent to a LiteWing ESP32-S3 drone over Wi-Fi using the Crazyflie client library (cflib); in the current bench test a validated activation triggers a motor-output pulse limited to 1.0 s with all propellers removed. A 60-second recording shows a stable resting baseline and discrete activation bursts in the RMS envelope. Latency, accuracy and flight results have not yet been measured; this paper reports the implemented pipeline and preliminary observations.

Keywords— *Electromyography, EMG signal processing, muscle-controlled drone, human–machine interaction, NPG Lite, LiteWing ESP32-S3, RMS activation detection, assistive technology.*

I. INTRODUCTION

Drones are widely used in photography, agriculture, surveillance and research, and are normally flown with a handheld radio controller, joystick or smartphone application. These interfaces require continuous manual manipulation of physical or on-screen controls, which demands practice from beginners and may be difficult for users who cannot operate conventional controllers comfortably.

Surface EMG records the electrical activity produced when skeletal muscles contract [3]. Because a deliberate contraction raises the amplitude of the signal, even simple amplitude-based detection can act as a switch, and earlier work has shown that forearm EMG can steer a quadcopter [1], [2]. EMG therefore offers a hands-free alternative to manual controllers and a practical teaching platform that joins biomedical sensing, embedded systems and wireless communication.

NeuroFlight is a prototype that connects a two-channel forearm EMG interface to a LiteWing ESP32-S3 drone. The main contributions of the current work are: (i) acquisition of left and right forearm EMG from an NPG Lite device over BLE; (ii) real-time causal filtering and RMS-based activation detection with per-session calibration; (iii) a safety-oriented command layer using hysteresis, latching and a pulse-time limit that drives the drone's motor outputs over Wi-Fi in a propeller-free bench test; and (iv) a live dashboard showing signals, classified command, connection state and motor state.

II. PROBLEM STATEMENT

Remote controllers and mobile applications give direct control but need both hands and learned coordination, and incorrect inputs can cause unwanted drone movement. An EMG-based alternative must overcome several difficulties: EMG amplitude varies between users and with electrode placement, skin condition and fatigue; the signal is easily corrupted by motion artefacts and 50 Hz power-line interference; and a false activation can translate directly into unintended motor action. The problem addressed here is to build an affordable system that acquires forearm EMG, rejects noise, detects intentional contractions reliably, and converts them into drone commands with built-in safety limits.

The present implementation is deliberately limited. It uses one user, two channels (left and right forearm), threshold-based detection rather than machine learning, and laptop-side processing. Directional flight and controlled yaw are not implemented; the drone is driven only through direct motor-output parameters during a bench test with propellers removed, and no flight tests have been performed.

III. LITERATURE SURVEY

Ali et al. controlled a quadcopter with EMG collected from arm muscles, detecting hand gestures, filtering the signals and classifying them into drone commands [1]. Miehlbradt et al. developed a data-driven body–machine interface in which natural upper-body movement and muscle activity steer a drone, presenting it as an easier-to-learn alternative to joystick control [2]. These studies motivate the NeuroFlight approach of mapping forearm muscle activity to flight commands.

Raez et al. reviewed EMG analysis, covering signal detection, processing, classification and applications [3], and De Luca

discussed practical recommendations for surface EMG recording, including the importance of electrode placement and the usable frequency content of the signal [4]. Based on this background, the project uses band-pass filtering, notch filtering for power-line interference, and an RMS envelope as a robust amplitude feature, leaving trained classifiers for future work.

IV. SYSTEM ANALYSIS

A. Proposed Architecture

The system has a sensing layer, a processing layer, a communication layer, an actuation layer and a presentation layer. Surface electrodes on the left and right forearms feed the NPG Lite, which streams samples over BLE. A Python program on a laptop filters the signals, detects activation, applies the safety logic and writes motor parameters to the LiteWing ESP32-S3 drone over Wi-Fi. A Tkinter dashboard shows the live state.

Forearm electrodes (L/R) → NPG Lite → BLE, 500 Hz
Laptop (Python): DC/band-pass/notch → RMS → threshold → command
cflib over Wi-Fi (UDP) → LiteWing ESP32-S3 motor outputs
Live dashboard: signals, command, connections, motor state

Fig. 1. End-to-end workflow of the proposed NeuroFlight system.

B. Hardware and Software

Hardware: NPG Lite two-channel biosignal acquisition device with disposable surface electrodes; LiteWing ESP32-S3 drone (Wi-Fi, four brushless motors); laptop computer. **Software:** Python 3.11 with *bleak* for BLE, *cflib* (Crazyflie client library) for drone communication, NumPy and SciPy for filtering [5], and Tkinter for the dashboard; the environment is managed with *uv*. The onboard IMU, time-of-flight and optical-flow sensors of the drone are not used in the present bench test.



Fig.2 SYSTEM ARCHITECTURE

C. Main Modules

The **acquisition module** connects to the NPG Lite by its BLE address, subscribes to the data characteristic, writes a START command, and reassembles notifications into 7-byte samples consisting of a 1-byte counter and three 16-bit big-endian channel values; channels 0 and 1 are used as the left and right forearm signals. The **conditioning module** applies a streaming fourth-order Butterworth band-pass filter (20–100 Hz, second-order sections) followed by a 50 Hz notch filter (quality factor 30) to each channel, preserving filter state between samples.

The **detection module** performs calibration, computes a sliding-window RMS envelope and compares it with per-channel thresholds. The **command and safety module** converts detections into the commands LEFT, RIGHT or STOP and gates motor output. The **drone interface** runs in a dedicated worker thread fed by a queue; it sets the *motorPowerSet* parameters (m1–m4 and enable) through cflib and always zeroes and disables the outputs on stop, error, disconnect or shutdown. The **dashboard** receives state updates through a queue so that the user interface never blocks signal processing.

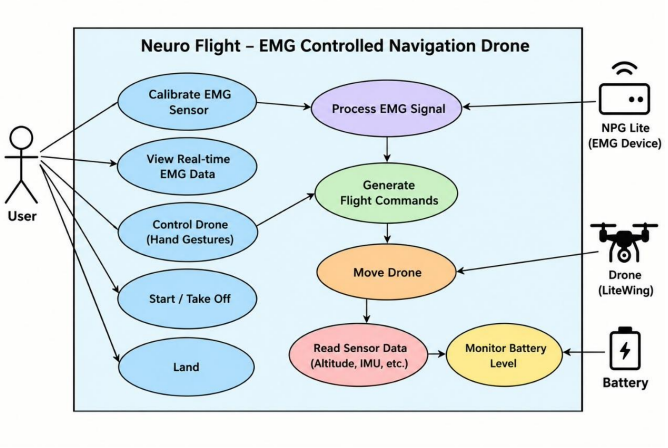


Fig.3 USE CASE DIAGRAM

V. METHODOLOGY

First, the program connects to the LiteWing drone (15 s timeout) and forces all motor outputs to zero, then connects to the NPG Lite over BLE (20 s timeout) and starts the EMG stream. Second, during the first three seconds the user keeps both forearms relaxed while filtered samples are collected for calibration. For each channel c the baseline b_c is the median of the calibration samples, the resting noise is

$$\sigma_c = \sqrt{(1/N)\,\Sigma\,(x_i - b_c)^2}, \quad (1)$$

and the activation threshold is

$$T_c = \max(k\cdot\sigma_c,\ \sigma_c + 20),\ k = 2.5. \quad (2)$$

Third, in live monitoring each filtered sample updates a sliding window of W = 25 samples (50 ms at 500 Hz) and the RMS envelope

$$RMS_c[n] = \sqrt{(1/W)\,\Sigma\,(x_c[i] - b_c)^2},\ i = n-W+1 \ldots n \quad (3)$$

is compared with T_c. A channel is active when its RMS is at or above its threshold; the system is considered released only when both channels fall below 0.65 × their thresholds, which provides hysteresis. Fourth, the classified state is applied as in Table I.

TABLE I. COMMAND CLASSIFICATION

Condition	Displayed command / action
Left only active	LEFT EMG ACTIVATED; one bench pulse
Right only active	RIGHT EMG ACTIVATED; one bench pulse
Both active	BOTH ACTIVE — STOP; outputs off
Neither active	NO MUSCLE ACTIVATION; idle

Fifth, safety rules govern the output. A single-channel activation starts one pulse and is latched until both channels are released, so holding a contraction cannot retrigger the motors; each pulse is limited to 1.0 s; simultaneous activation of both channels, any processing exception, a disconnect or window close immediately requests motor stop. During the bench test all four motors are set to a test power of 1000 through the drone's parameter interface. The parameters of the implementation are listed in Table II.

TABLE II. SYSTEM PARAMETERS

Parameter	Value
Sampling rate	500 Hz
Sample packet	7 bytes (counter + 3 × 16-bit)
Band-pass filter	Butterworth, 4th order, 20–100 Hz
Notch filter	50 Hz, Q = 30
RMS window	25 samples (50 ms)
Calibration	3 s, relaxed
Threshold	max(2.5σ, σ + 20)
Release ratio	0.65 of threshold
Motor test power	1000 (propellers removed)
Max pulse length	1.0 s

Each sample costs a fixed number of operations, so processing time per sample is constant, and memory use is constant because only the filter states and a 25-sample window are kept. In the present code the dominant delay is the 0.15 s pause after every parameter write: switching the outputs on takes six writes (about 0.9 s) and switching them off takes five (about 0.75 s), which is comparable to the 1.0 s pulse limit.

VI. RESULTS AND DISCUSSION

The implemented pipeline acquires two-channel EMG, filters it, computes the RMS envelope, classifies activation and displays the result on the dashboard (Fig. 4). The dashboard in the captured state shows the NPG Lite as waiting, the LiteWing link as connecting and a LiteWing connection timeout message, which illustrates that the end-to-end BLE and Wi-Fi bench verification is still in progress.

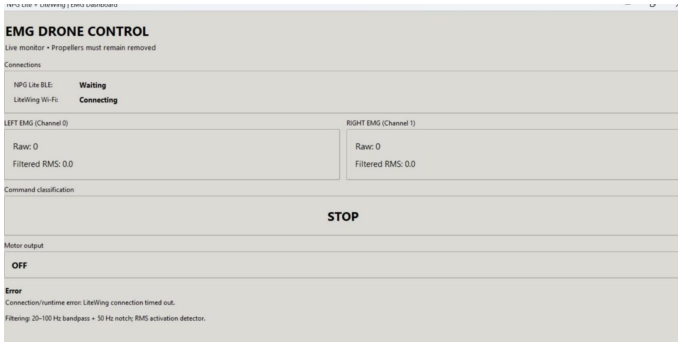


Fig.4 DASHBOARD OVERVIEW

Fig. 5 shows a 60-second raw two-channel recording. The resting baseline sits at about 1600 ADC counts. During the first three seconds both channels show large excursions, with samples reaching the converter limits, which is a start-up transient rather than muscle activity. After this the baseline is stable and short bursts of larger amplitude appear.

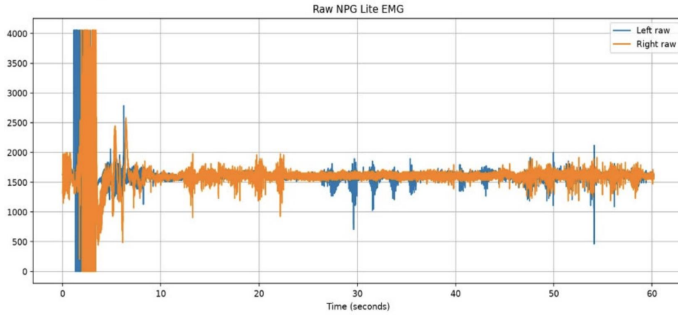


Fig.5 RAW NPG LITE EMG

After band-pass and notch filtering (Fig. 6) the baseline offset is removed and the signal is centred on zero. Apart from the start-up transient, which reaches about ±2000 counts, the resting signal stays within a few hundred counts and individual bursts stand out clearly.

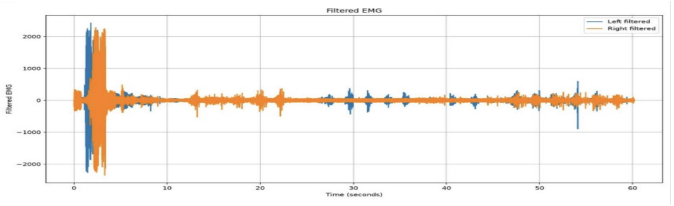


Fig.6 FILTERED EMG

The RMS envelope (Fig. 7) turns these bursts into smooth peaks. Beyond the start-up transient, which peaks near 1750, activations appear as isolated peaks of roughly 100–300 counts at several instants between about 13 s and 54 s, with the left and right channels distinguishable. These values are read from the plotted curves and are approximate.

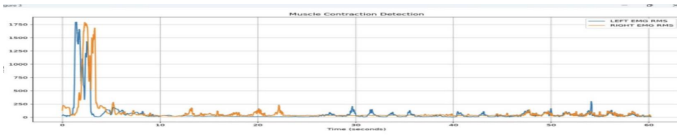


Fig.7 MUSCLE CONTRACTION DETECTION (RMS ENVELOPE)

TABLE III. IMPLEMENTATION STATUS

Component	Status
EMG acquisition and CSV recording	Completed
Offline filtering and RMS	Completed
Command logic	Completed
Dashboard (sample data)	Completed
NPG Lite BLE live link	In progress
LiteWing Wi-Fi link	In progress
Motor response verification	Pending
Flight testing	Not performed

Limitations. Quantitative measures such as response time, CPU and memory use, classification accuracy and false-trigger rate have not yet been measured, and the planned test cases for noise reduction and contraction detection are still pending. The start-up transient overlaps the three-second calibration window, which can inflate the learned thresholds, and a single user and a single recording were used. The system should therefore be regarded as a working signal pipeline and safe bench demonstrator rather than a validated flight controller.

VII. CONCLUSION

NeuroFlight demonstrates a low-cost route from forearm muscle activity to drone motor commands. It combines BLE acquisition from an NPG Lite, causal band-pass and notch filtering, RMS-based activation detection with per-session calibration, a hysteresis and pulse-limited safety layer, and a Wi-Fi link to a LiteWing ESP32-S3 drone, together with a live dashboard. The recorded signals show clear separation between rest and activation bursts after filtering. Because latency, accuracy and flight performance remain to be measured, the present results should be taken as evidence of a functioning pipeline and not as proof of reliable flight control.

VIII. FUTURE SCOPE

Planned improvements include priming the filter state and discarding the start-up transient before calibration; reducing parameter-write delay so that stop commands act faster; measuring end-to-end latency, detection accuracy and false triggers across several users; and adding more gestures with a trained classifier to give directional commands through flight setpoints rather than raw

motor power. Further work includes an explicit emergency stop, use of the drone's onboard IMU, time-of-flight and optical-flow sensors, moving signal processing onto the microcontroller, a calibration interface, and carefully supervised controlled flight tests.

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